PARKINSON'S TELEMONITORING ANALYSIS

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Abstract

The aim of the research: The aim of this study is to investigate the extent to which, and the accuracy with which, the severity of motor symptoms in Parkinson's disease can be estimated based on the *motor_UPDRS* score using voice-based acoustic biomarkers. The study also aims to determine whether a nonlinear machine learning model provides a predictive advantage over classical linear regression.

Research questions:

- To what extent can the variance in *motor_UPDRS* be explained using acoustic voice parameters?
- Does the Random Forest regression model significantly improve prediction accuracy compared to linear regression?
- Which acoustic voice parameters have the greatest predictive importance in estimating *motor_UPDRS*?

Data: Oxford Parkinson's Disease Dataset

Methods: *Exploratory Data Analysis (EDA), correlation analysis, row-level 80/20 train-test split, linear regression, Random Forest regression, RMSE/R^2-based model evaluation*, and *feature importance analysis*.

Main Findings: Based on the analysis, voice-based biomarkers appear to be suitable to some extent for estimating the severity of motor symptoms in Parkinson's disease; however, their predictive capability remained limited. The *Random Forest model* identified certain patterns between the acoustic variables and *motor_UPDRS* values, but based on the low R^2 and relatively high *RMSE* measured on the test set, the accuracy of individual-level predictions remained limited. According to the feature importance analysis, several voice parameters, such as *PPE*, *shimmer*- and *jitter*-based measures, and *HNR* played a more important role in prediction than other variables.

Neurological background

Parkinson’s disease:

Parkinson’s disease is one of the most common adult-onset neurodegenerative disorders, with a chronic and progressive course. The disease primarily involves disturbances in motor control; however, it is not solely a motor disorder, as a wide range of non-motor symptoms may also appear in the clinical presentation. The disease was first described by James Parkinson in 1817 in his work Essay on the Shaking Palsy, while an important milestone in the pathological understanding of the disease was provided by Frederick Lewy, who reported in 1912 the discovery of intracellular inclusions associated with the disorder, the so-called Lewy bodies, which are still considered one of the key neuropathological features of the disease today.

The underlying pathology of the disease is primarily the gradual degeneration of dopamine-producing neurons, particularly in the *substantia nigra pars compacta*. As a result, dopamine levels decrease in the striatum, disrupting the function of the basal ganglia, which are responsible for motor control. Dopamine does not directly initiate movement; rather, it plays a role in maintaining the balance between movement initiation and movement inhibition. In simpler terms, dopamine supports the movement-promoting *“GO”* system while reducing the activity of the movement-inhibiting *“STOP”* system.

Under normal functioning, the basal ganglia help determine which movements should be initiated and which should be inhibited. The thalamus transmits information to the motor cortex, which then issues the specific motor commands, while the cerebellum is involved in fine-tuning movement, coordination, balance, and precise execution.

In Parkinson’s disease, this system loses its balance due to dopamine deficiency. The direct pathway, which supports movement initiation, becomes less effective, while the indirect pathway, which is involved in movement inhibition, becomes overactive. As a result, the output nuclei of the basal ganglia exert an increased inhibitory effect on the thalamus, so fewer excitatory signals reach the motor cortex. This process contributes to the slowing of movements and the difficulty of their execution.

The classic motor symptoms of Parkinson’s disease include *bradykinesia*, meaning the slowing of movements, resting *tremor*, *rigidity*, meaning muscle stiffness, and postural instability and *balance impairment*, which tend to appear in later stages. These symptoms are directly related to damage to the dopaminergic system and dysfunction of the basal ganglia.

It is important to note that the symptoms of the disease do not necessarily appear immediately. In the early stages of the disease, the brain attempts to compensate for dopamine deficiency through various compensatory mechanisms. These may include changes in the sensitivity or number of dopamine receptors, increased activity of the remaining dopamine-producing neurons, reduced breakdown of dopamine, and reorganization of neural networks.

For this reason, Parkinson's disease may remain asymptomatic for a long time or present only with mild symptoms that are difficult to recognize. The characteristic motor symptoms usually become clinically visible when these compensatory mechanisms are no longer sufficient.

Parkinson's disease is not associated only with motor symptoms. In the prodromal stage of the disease, meaning the phase preceding the classic motor symptoms, reduced sense of smell, loss of smell, constipation, REM sleep behavior disorder, depression, anxiety, or other mood disturbances may often occur.

As the disease progresses, non-dopaminergic systems may also become impaired, such as the noradrenergic, serotonergic, and cholinergic pathways. This explains why Parkinson's disease may also be associated with psychiatric symptoms, autonomic dysfunction, sleep disorders, and cognitive decline. The combined presence of motor and non-motor symptoms can significantly reduce the quality of life of affected individuals.

From an epidemiological perspective, Parkinson's disease represents a significant public health problem worldwide. Based on the provided data, the global prevalence is approximately 129 per 100,000 people, while the annual incidence is approximately 108–212 per 100,000 people.

The number of cases shows an increasing trend over time: in 2016, approximately 6.1 million people with Parkinson's disease were recorded, while in 2021 this number reached 11.8 million. By 2025, some estimates project an even higher number of patients.

This increase may be partly explained by the ageing population; however, the age-adjusted increase suggests that the trend is unlikely to be explained solely by population ageing. Possible contributing factors may include environmental influences, lifestyle changes, improvements in diagnostics, and the interaction between genetic and environmental factors.

From the perspective of the present project, it is particularly important that Parkinson's disease can affect not only the movement of the limbs, but also speech and voice production. Voice production is a complex motor process involving respiration, the function of the laryngeal muscles, articulation, and fine motor coordination.

Since impaired motor control in Parkinson's disease may also affect speech motor function, voice-based acoustic features may carry potential information about disease severity. This justifies the investigation of variables such as *jitter*, *shimmer*, *HNR*, and *PPE*, which provide insight into frequency variation, amplitude variation, noisiness, and the overall quality of the voice.

The significance of measurement:

Quantitative monitoring of Parkinson's disease symptoms is of particular importance, as the course of the disease varies between individuals and the patient's condition may change over time. The *UPDRS* is a widely used, standardized clinical scale that enables the structured assessment of the severity of both motor and non-motor symptoms.

The *motor_UPDRS* score describes the severity of motor symptoms. The advantage of telemonitoring data is that, through repeated measurements, they can track changes in a patient's condition and may therefore provide a more detailed picture of the progression of the disease over time.

Data-driven approach:

The aim of data-driven approaches is to complement clinical assessment with objectively measurable biomarkers, such as voice-based acoustic features. Since Parkinson's disease may also affect speech and voice production, variables such as *jitter, shimmer, HNR,* and *PPE* may potentially carry information about the severity of the motor condition.

Machine learning models can be used to investigate the extent to which the *motor_UPDRS* score can be estimated from these voice parameters, and whether acoustic biomarkers may be suitable for supporting clinical monitoring.

Hypotheses

- Voice-based biomarkers are able to significantly estimate *motor_UPDRS*.
- Nonlinear machine learning models provide better predictive performance than linear regression.
- In the estimation of *motor_UPDRS*, some acoustic biomarkers have greater predictive importance than other variables.

Dataset description

Sample size: 5,875 voice samples / measurement records from patients with early-stage Parkinson's disease.

Variables: Demographic data, voice features, and motor and total *UPDRS* scores.

Key variable: *motor_UPDRS*.

Missing data: There were no missing values / no missing data were identified.

Type of measurement: Regular voice samples recorded in a home environment.

Methods

The project is based on the Parkinson's Telemonitoring Dataset, which contains 5,875 voice recording-based observations from female and male patients with early-stage Parkinson's disease. One observation represents a measurement record captured from a voice sample belonging to a given patient. In addition, the dataset includes demographic variables, voice-based acoustic features, and clinical severity measures.

The main objective of the analysis was to estimate the severity of motor symptoms associated with Parkinson's disease. Accordingly, the target variable was the *motor_UPDRS* score. A higher *motor_UPDRS* value indicates more severe motor symptoms, while a lower value indicates less severe symptoms.

Therefore, the project was not formulated as a classification task, meaning it did not aim to distinguish between patients with Parkinson's disease and control subjects. Instead, it was defined as a regression problem, where the goal was to predict the *motor_UPDRS* score based on demographic and voice-based variables.

The analysis was carried out in the *R* environment using *Microsoft Visual Studio Code*. For data handling and visualization, the *tidyverse package* was used; for the initial overview of the data, the *skimr package* was applied; for the *Random Forest model*, the *randomForest package* was used; for examining multicollinearity, the *car package* was applied; and for additional figure formatting, the *ggpubr package* was used.

Before the modeling stage, the dataset was split into training and test sets. Before the modeling stage, the dataset was split into training and test sets. After importing the data, the column names were standardized using the *R make.names()* function so that the variable names could be safely used in the subsequent analyses and modeling.

During data preparation, the structure of the dataset was reviewed, descriptive statistics were calculated, and missing values were checked. Following this, the analysis focused on the most important variables.

In the models, the variables *age*, *sex*, *HNR*, *PPE*, *Shimmer.APQ11*, and *Jitter.RAP* were included as predictors. Among these, ageand sex are demographic characteristics, while the other variables describe different aspects of voice quality, frequency variation, amplitude variation, and voice clarity.

During the exploratory data analysis, I first examined the distribution of the target variable, namely *motor_UPDRS*. For this purpose, a histogram and a boxplot were created, which made it possible to review the general distribution of motor symptom severity.

As part of the *EDA*, scatter plots were used to examine the relationship between age, selected voice features, and the *motor_UPDRS* score. The sex variable was included as a predictor in the models.

Before the modeling stage, the dataset was split into training and test sets. The split was performed at the row level: 80% of the data were randomly assigned to the training set, while the remaining 20% were assigned to the test set. This approach made it possible to evaluate model performance on data that were not directly used during training.

In addition, two regression models were developed in the project. First, a *linear regression* model was used as an initial, easily interpretable baseline model. This was followed by the application of a *Random Forest* regression model, which may be more suitable for identifying nonlinear relationships and more complex variable patterns.

Both models used the same predictors: *PPE, Shimmer.APQ11, Jitter.RAP, HNR, age,* and *sex*. In the case of the linear model, multicollinearity was also assessed using *VIF* values in order to examine whether strong associations existed among the explanatory variables.

The performance of the *Random Forest* model was measured on the test set, while in the case of the linear regression model, the summary statistics and *RMSE* value of the model fitted on the full dataset were interpreted. Two main metrics were used for evaluation: *RMSE* and R^2.

RMSE indicated the average extent to which the model's predictions differed from the actual *motor_UPDRS* values. R^2, on the other hand, showed how much of the variance in the target variable could be explained by the model.

Results

No missing data were identified in the dataset.

The *motor_UPDRS* values ranged approximately between 5 and 40, with a mean of around 21.3 and a median of around 20.9. This suggests that a substantial proportion of the observations were concentrated around moderate motor symptom severity, while both milder and more severe cases were also present in the data.

A positive relationship was observed between age and *motor_UPDRS*: older patients generally showed higher motor symptom severity. In contrast, the relationship between *HNR* and *motor_UPDRS* was weakly negative, which may suggest that higher *HNR* values, indicating clearer voice quality, may be associated with lower *motor_UPDRS* scores.

However, given the substantial variability in the data, these relationships alone cannot be considered sufficient to accurately explain motor symptom severity.

The performance of the *Random Forest* model was limited: the R^2 value was approximately *0.024*, while the *RMSE* was approximately 9.03. This indicates that the model was able to explain only a small proportion of the variability in *motor_UPDRS* scores, and that the accuracy of individual-level predictions remained limited.

A feature importance analysis was also performed for the *Random Forest model*. Based on its results, age proved to be the strongest predictor. This was followed by the voice-based features, particularly *HNR, PPE, Shimmer.APQ11,* and *Jitter.RAP*. The role of the sex variable in the model did not appear to be as substantial.

Based on the results, demographic and voice-based variables contain some information about the severity of motor symptoms; however, the selected predictors alone did not prove sufficient for accurately predicting the *motor_UPDRS* score.

Overall, the aim of the methodology was to highlight the relationship between voice-based and demographic characteristics of patients with Parkinson's disease and the severity of motor symptoms. The *EDA* helped to understand the structure of the data and its main patterns, while linear regression and the *Random Forest model* provided an opportunity to examine the predictive estimation of the *motor_UPDRS* score.

Based on the results, the model was able to identify certain general trends, but its predictive performance remained limited. Therefore, the approach is more exploratory in nature rather than a model directly suitable for clinical prediction.

1.1Explained Variance

In the case of the *Random Forest model*, the R^2 value was approximately 0.024, which means that the model was able to explain only a small proportion of the variance in *motor_UPDRS* scores. Based on this, the selected demographic and voice-based predictors had only limited explanatory power on their own.

However, it should be noted that some variables, particularly age and certain voice features, showed a relationship with motor symptom severity, but these were not sufficient for accurately predicting *motor_UPDRS* values.

1.2 Machine Learning Model Performance

In the present analysis, the machine learning approach was represented by the *Random Forest regression model*. The aim of the model was to examine whether a nonlinear method could improve the prediction of the *motor_UPDRS* score compared to linear regression. A gradient boosting model was not applied in the analysis; therefore, the comparison can only be interpreted between linear regression and the *Random Forest* model.

Based on the results, the performance of the *Random Forest model* remained limited. The R^2 value measured on the test set was approximately 0.024, while the *RMSE* was approximately 9.03. This indicates that the model was able to explain only a small proportion of the variability in *motor_UPDRS* scores and did not show sufficiently strong predictive performance to support a clear clinical improvement compared to linear regression.

Clinical Interpretation

From the perspective of clinical interpretation, the most important finding of the analysis is that voice-based acoustic features and demographic variables can be associated to some extent with the severity of motor symptoms in Parkinson's disease; however, on their own, they did not prove sufficient for accurately predicting the *motor_UPDRS* score.

This means that the examined variables carry clinically interpretable information, but based on the model's performance, they should be interpreted more as complementary biomarkers rather than as an independent diagnostic or decision-support tool.

The *motor_UPDRS* score is used to measure the severity of motor symptoms in Parkinson's disease. A higher value indicates a more severe motor condition; therefore, the aim of the model was not merely statistical prediction, but also to examine the extent to which features extracted from voice samples may reflect patients' movement-related clinical status.

This is particularly important because, in Parkinson's disease, changes in speech and voice quality may often be associated with disease progression, such as the voice becoming softer, more tremulous, less stable, or noisier.

Based on the results, age proved to be one of the strongest predictors. This is also clinically interpretable, as the occurrence and severity of Parkinson's disease are often associated with advancing age.

Older patients generally showed higher *motor_UPDRS* values, which suggests that age may be an important background factor in estimating motor symptom severity. However, age alone does not fully explain disease severity, as the progression of Parkinson's disease can vary considerably between individuals.

Among the voice-based variables, *HNR, PPE*, *Shimmer.APQ11*, and *Jitter.RAP* played a particularly notable role. *HNR* is a measure related to voice clarity and the ratio of harmonic to noisy components.

The weakly negative relationship between *HNR* and *motor_UPDRS* may suggest that higher *HNR* values, indicating clearer voice quality, may be associated with lower motor symptom severity. This is clinically interpretable, as speech may often become less stable, softer, or noisier as Parkinson's disease progresses.

PPE, shimmer-type, and *jitter-type* variables describe frequency and amplitude variations in the voice. These measures may be relevant because Parkinson's disease can affect not only limb movements, but also the fine motor control required for speech production.

If voice production becomes less stable, this may be reflected in frequency variation, amplitude changes, or a deterioration in voice periodicity. Based on this, voice-based biomarkers may potentially be suitable for providing complementary information about the patient's condition.

At the same time, the results should be interpreted with caution based on the model's performance. The R^2 value of the *Random Forest model* was approximately 0.024, while the *RMSE* was approximately 9.03. This means that the model was able to explain only a very small proportion of the variance in *motor_UPDRS* scores, while the average prediction error can be considered relatively large from a clinical perspective.

A difference of approximately 9 points on the *motor_UPDRS* scale may already represent a substantial difference in the assessment of a patient's condition. Therefore, in its current form, the model is not suitable for accurate patient-level clinical prediction.

Based on the comparison between the linear regression model and the *Random Forest* model, it cannot be stated with certainty that the nonlinear machine learning approach resulted in a clearly clinically useful improvement. Although *Random Forest* may be more suitable for identifying more complex, nonlinear relationships, the performance measured on the test set indicated that the model's generalization ability remained limited.

This suggests that the selected predictors did not contain sufficient information for accurately estimating motor symptom severity, or that the development of *motor_UPDRS* is also influenced by additional factors that were not included in the model.

From a clinical perspective, the results are primarily exploratory in nature. The study indicates that voice-based acoustic features may be associated with the motor symptoms of Parkinson's disease, but the relationship is not strong enough for the model to be used independently for diagnosis, disease monitoring, or supporting therapeutic decisions.

Voice-based measurement should rather be interpreted as a potential complementary tool that, together with other clinical data, may help provide a better understanding of the patient's condition.

From the perspective of practical application, an advantage of voice-based telemonitoring is that it is non-invasive, relatively easy to repeat, and may even be used in a home environment. In the longer term, this could be useful for monitoring patients' conditions, particularly if the analysis of voice samples is combined with other data, such as medication information, disease duration, clinical examination results, or movement-based sensor data.

However, the present analysis rather indicates that voice-based data alone are not yet sufficient for developing a reliable clinical prediction system.

Overall, the results indicate that demographic and voice-based variables contain some clinically interpretable information about the motor severity of Parkinson's disease, but the predictive performance remained limited.

Therefore, in its current form, the model cannot be considered a clinical decision-support tool, but rather a data-driven, exploratory analysis that may contribute to a better understanding of the role of voice-based biomarkers in the study of Parkinson's disease.

Limitations and Future Directions

One important limitation of the analysis is that although the dataset contains 5,875 observations, these do not come from 5,875 different patients; rather, multiple measurements may belong to the same patients. Therefore, the sample size should be interpreted with caution: the number of observations is relatively high, but the actual number of patients is considerably lower.

This may limit the generalizability of the results to larger and more heterogeneous populations of patients with Parkinson's disease. As a result, the findings should primarily be interpreted in relation to the patients and measurement conditions represented in the given dataset. Another limitation may be measurement bias arising from the recording of voice samples. Voice-based variables may be influenced by microphone quality, background noise, the patient's current level of fatigue, voice intensity, the time of day, and the environment in which the measurement was taken.

In addition, the symptoms of Parkinson's disease may fluctuate over time; for example, motor or speech-related states may differ before and after medication intake. These factors may introduce noise into the data and reduce the predictive performance of the model.

From the perspective of generalizability, it is also important that the dataset consists of patients with early-stage Parkinson's disease. For this reason, the results may not necessarily be applicable to patients in more advanced disease stages, or to other age, language, or clinical groups.

Since the analysis does not include a healthy control group, the model was not developed for diagnostic purposes, but rather to estimate motor symptom severity in patients already diagnosed with Parkinson's disease. Therefore, the results are primarily exploratory in nature and should not be interpreted as a direct clinical decision-support system.

One important direction for future development could be the more detailed incorporation of longer-term longitudinal data. Since multiple measurements from different time points are available for the patients, it would be worthwhile to apply models that are specifically capable of handling temporal changes and within-subject differences.

In addition, it would be useful to include further clinical variables, such as disease duration, medication status, treatment phase, or other measures describing motor and non-motor symptoms.

As a further opportunity, the application of deep learning techniques could also be investigated, especially if raw voice recordings or more detailed acoustic features are available. Neural networks may be capable of identifying more complex, nonlinear patterns; however, this would require a larger and more diverse dataset, as well as rigorous validation.

Overall, the present analysis provides a useful starting point for the investigation of voice-based biomarkers, but further data, more advanced modeling approaches, and external validation would be needed for clinical applicability.

Preferences

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